

J.R. Gutierrez

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SUMMARY

Product Security Engineer Lead with 25+ years of software development, 12+ years of cyber and software testing, and 6+ years of technical leadership experience. Currently leading security test and verification efforts for B-52 avionics systems at Boeing, with specialized expertise in RF security testing, Anti-Tamper verification, and automated test tool development. Active Top Secret clearance with proven track record in leading cyber test events, developing enterprise test capabilities, and mentoring engineering teams.

EXPERIENCE

Product Security AT Software Lead

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The Boeing Company

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May 2019 - Present

- Active Top-Secret Clearance with Special Access Program (SAP) authorizations
- Lead cyber test and verification efforts for B-52 avionics systems, specifically the radar display subsystem, ensuring compliance with DoD security requirements and Anti-Tamper certification standards
- Serve on Boeing's Anti-Tamper Technical Board, reviewing and approving verification strategies for AT solutions across all enterprise programs
- Develop and maintain comprehensive test documentation including test plans, procedures, and reports for avionics security testing
- Architect and implement automated test tools and CI/CD DevSecOps pipeline utilizing Python, GoogleTest, and FFF (Fake Function Framework) for continuous security verification
- Lead technical team of up to 7 software engineers, providing code reviews, architecture decisions, and task delegation while mentoring in secure software development and test methodologies
- Conduct security assessments including vulnerability analysis, penetration testing, and risk mitigation for mission-critical aircraft systems
- Develop secure embedded C software (50% hands-on development) for Anti-Tamper integrations, adhering to NIST SP 800-53 and DoD security standards
- Coordinate with cross-functional teams, suppliers, and government agencies to resolve complex integration issues and enhance security standards
- Represent security testing capabilities in engagements with Boeing, government, and customer leadership, providing technical recommendations on security risks and solutions

Cybersecurity IAM Analyst II

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Sonic Drive-In Corporation

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Feb 2019 - May 2019

- Conducted penetration testing using Active Directory security tools (BloodHound) to identify privilege escalation paths and critical vulnerabilities across enterprise infrastructure
- Architected a secure privileged access management (PAM) system across 2,000+ endpoints, utilizing containerized microservices to ensure zero-downtime deployment

- Developed test methodologies and documentation for security validation of identity and access management systems

Security Ops Engineer Analyst

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American Fidelity Assurance Company

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Jun 2018 - Jan 2019

- Automated vulnerability scans and incident response processes using C/C++, Python, and PowerShell scripts integrated with a SOAR platform, reducing response times and enhancing security posture
- Conducted threat analysis and remediation, applying systems thinking to anticipate and mitigate risks across interconnected systems

Senior Security Engineer

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Seagate Technology

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Mar 2013 - Mar 2018

- Built automated security testing tool for SOX compliance that cross-referenced system access lists with employee authorization status, streamlining verification processes
- Led cross-functional teams of junior security engineers in security integration projects, overseeing full-stack security testing from Linux OS to application layers
- Conducted penetration testing and security assessments on financial systems to ensure SOX compliance and identify vulnerabilities
- Developed comprehensive security architectures based on NIST CSF and ISO 27001 frameworks, ensuring compliance for mission-critical data center infrastructure
- Defined security test requirements for supplier integrations, ensuring secure component incorporation into complex systems

Owner & Security Consultant

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9Seats Inc.

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Jan 2008 - Sep 2011

- Conducted RF security testing and penetration testing for wireless systems including WiFi (DoS, deauth attacks, key cracking), ZWave, Zigbee protocols using Software Defined Radio (SDR), and RFID/NFC security assessments
- Developed mobile applications and web systems with integrated security testing throughout the development lifecycle
- Led technical teams in outsourced software development for international clients (India, Ukraine)
- Designed secure software systems and vendor security requirements for 20+ client organizations

LEADERSHIP

President and Chairman

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(ISC)2 Oklahoma City Metro Chapter

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March 2020 - March 2023

- Launched and led the chapter, increasing member engagement and fostering a proactive cybersecurity community
- Developed a mentorship program to guide aspiring professionals, enhancing industry talent pipelines

EDUCATION

B.S. in Information Technology, Security Emphasis

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March 2019

Western Governors University | Salt Lake City, UT

SKILLS

Programming Languages:

C/C++, Ada, Python, Bash, PowerShell, Perl

Security Testing & Verification:

Penetration testing, vulnerability assessments, V&V processes, test automation, CI/CD DevSecOps pipelines, GoogleTest, FFF (Fake Function Framework)

RF Security Testing:

WiFi security (DoS, deauth, key cracking), Software Defined Radio (SDR), ZWave/Zigbee protocols, RFID/NFC security, Flipper Zero, wireless penetration testing

Security Tools & Platforms:

BloodHound, SOAR platforms, automated testing frameworks, vulnerability scanners

Security Architecture:

TPM/Secure Boot, key management, Red/Black separation, Zero Trust, Anti-Tamper

Security Frameworks:

NIST CSF, NIST SP 800-53, ISO 27001, DoD security standards

Cybersecurity Processes:

Incident response, threat analysis, vulnerability management, Anti-Tamper verification, encryption, penetration testing, Red Team operations

Engineering Disciplines:

Software engineering, systems engineering (avionics and defense systems), embedded systems, test engineering

CERTIFICATIONS

- CISSP - Certified Information Systems Security Professional
- CISSP-ISSEP - Information Systems Security Engineering Professional
- CISSP-ISSMP - Information Systems Security Management Professional
- CISSP-ISSAP - Information Systems Security Architecture Professional
- CEH - Certified Ethical Hacker
- CCNA - Certified Network Associate Routing and Switching
- CCNA Security - Cisco Certified Network Associate Security